



CMPG 325-COMPUTER NETWORKS

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CMPG 325 – COMPUTER NETWORKS

MILESTONE 2

NETWORK IMPLEMENTATION AND TESTING

Ipeleng Study & Tutoring Hub

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Introduction

This milestone delivers on the implementation of the network design that was proposed for the Ipeleng Study & Tutoring Hub. The implementation was done in Cisco Packet Tracer and utilizes VLAN segmentation, router-on-a-stick inter VLAN routing, DHCP, a management VLAN, and a guest wireless network.

The implementation meets the requirements described in Milestone 1: staff, student and visitor networks; VLAN segmentation; using the existing infrastructure without any civil works; and sufficient capacity to absorb roughly 25% growth without renumbering the Addressing Plan.

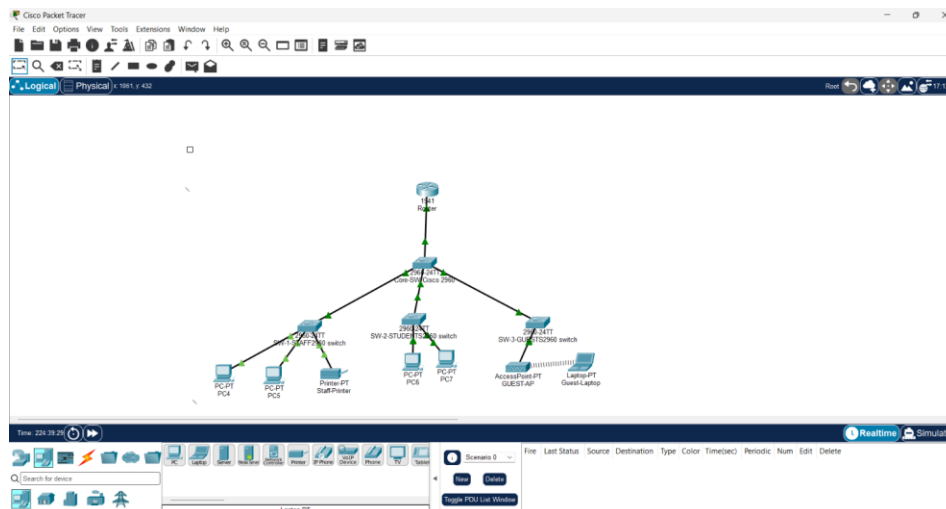
Client Requirements

- Isolate networks for staff, students, and visitors/guests.
- Segment traffic logically by using VLANs.
- Use the building's existing infrastructure and do not make civil works.
- Have enough addressing for the 25 percent expected growth.
- Keep the existing addressing scheme. No addressing renumbering is needed.

These needs would be met by assigning VLAN 10 for the Staff, VLAN 20 for the Students, VLAN 30 for Guests, and VLAN 99 for the network management.

Network Topology

A star topology has been configured. Cisco 1941 router is connected to the Core Switch while the Core Switch is connected to three (3) Cisco 2960 access switches. The Staff switch connects all the staff computers and the printer while the Student switch connects the student computers and the Guest switch connects the wireless access point.



Core-SW → SW-2-STUDENTS	Core-SW Fa0/1 → SW-2 Gi0/1
Core-SW → SW-3-GUESTS	Core-SW Fa0/2 → SW-3 Gi0/1
SW-1-STAFF → PC4/PC5/Printer	Fa0/1, Fa0/2, Fa0/3
SW-2-STUDENTS → PC6/PC7	Fa0/1, Fa0/2
SW-3-GUESTS → Access Point	Fa0/1 → Access Point Port 0
Access Point → Guest Laptop	Wireless connection

5. VLAN Configuration

VLAN	Name	Purpose
10	STAFF	Staff users and printer
20	STUDENTS	Student users
30	GUESTS	Guest wireless users
99	MANAGEMENT	Network-device management

VLANs were configured on the switches to define separate logical broadcast domains. The access ports were assigned to access lists, and the inter-switch links and uplinks to the router were configured as trunk links where appropriate.

6. IP Addressing Scheme

VLAN	Network	Subnet Mask	Default Gateway
10 – Staff	172.30.10.0/25	255.255.255.128	172.30.10.1
20 – Students	172.30.10.128/25	255.255.255.128	172.30.10.129
30 – Guests	172.30.11.0/26	255.255.255.192	172.30.11.1
99 – Management	172.30.11.64/27	255.255.255.224	172.30.11.65

The overall addressing block is 172.30.10.0/23. The subnet sizes provide separate address space for each user group while leaving capacity for future expansion.

7. Router-on-a-Stick Configuration

Inter-VLAN routing was implemented on the Cisco 1941 router using subinterfaces on GigabitEthernet0/0. IEEE 802.1Q encapsulation was configured for each VLAN.

Interface	VLAN	IP Address
Gi0/0.10	10	172.30.10.1
Gi0/0.20	20	172.30.10.129
Gi0/0.30	30	172.30.11.1
Gi0/0.99	99	172.30.11.65

The router interface verification showed the configured VLAN subinterfaces as operational during implementation.



8. Trunk Configuration

Trunking was configured between the Core Switch and the access switches, and between the Core Switch and router. Trunking carries tagged VLAN traffic between network devices while preserving VLAN separation.

[INSERT SCREENSHOT 4 HERE]

Figure 4. Trunk verification using show interfaces trunk.

9. DHCP Configuration

Pool	Network	Excluded/Reserved	Default Gateway
STAFF	172.30.10.0/25	172.30.10.1– 172.30.10.9	172.30.10.1
STUDENTS	172.30.10.128/25	172.30.10.129– 172.30.10.139	172.30.10.129
GUESTS	172.30.11.0/26	172.30.11.1– 172.30.11.9	172.30.11.1

The resulting intended client ranges are Staff 172.30.10.10–172.30.10.126, Students 172.30.10.140–172.30.10.254 and Guests 172.30.11.10–172.30.11.62. VLAN 99 management addresses were kept static.

```
Router
Physical Config CLI Attributes
IOS Command Line Interface
Excluded addresses : 3
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/
Excluded/Total
172.30.10.1 172.30.10.1 - 172.30.10.126 2 / 3
/ 126

Pool STUDENTS :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Excluded addresses : 3
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/
Excluded/Total
172.30.10.129 172.30.10.129 - 172.30.10.254 2 / 3
/ 126

Pool GUESTS :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 62
Leased addresses : 1
Excluded addresses : 3
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/
Excluded/Total
172.30.11.1 172.30.11.1 - 172.30.11.62 1 / 3
/ 62
Router#
```

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☐ Top

10. End-Device Configuration

Computers used by staff and students were configured to obtain their addresses via DHCP. This was evident in the implementation where PC4 was assigned 172.30.10.10/25 with gateway 172.30.10.1, while PC5 was assigned 172.30.10.11/25 with gateway 172.30.10.1.

A static IPv4 address of 172.30.10.5, and subnet mask of 255.255.255.128 and default gateway of 172.30.10.1 were configured on the Staff Printer.

PC4 DHCP configuration.

PC4

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 172.30.10.10

Subnet Mask 255.255.255.128

Default Gateway 172.30.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::203:E4FF:FE7E:200

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC5 DHCP configuration.

PC6

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 172.30.10.141

Subnet Mask 255.255.255.128

Default Gateway 172.30.10.129

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2E0:B0FF:FE14:2D87

Default Gateway

DNS Server

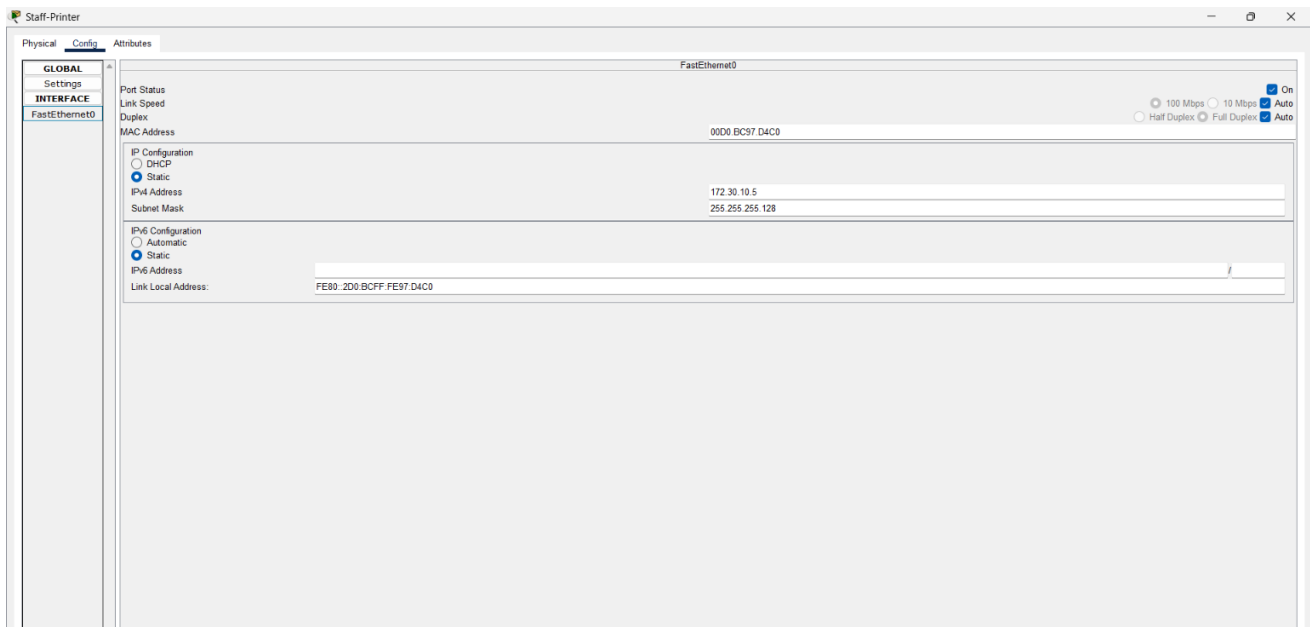
802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password



11. Guest Wireless Network

The guest wireless network was implemented using an Access Point-PT connected to SW-3-GUESTS. The wireless SSID was configured as IPEleng-Guest. The guest wireless client was configured to obtain its IPv4 address through DHCP on the Guest VLAN.

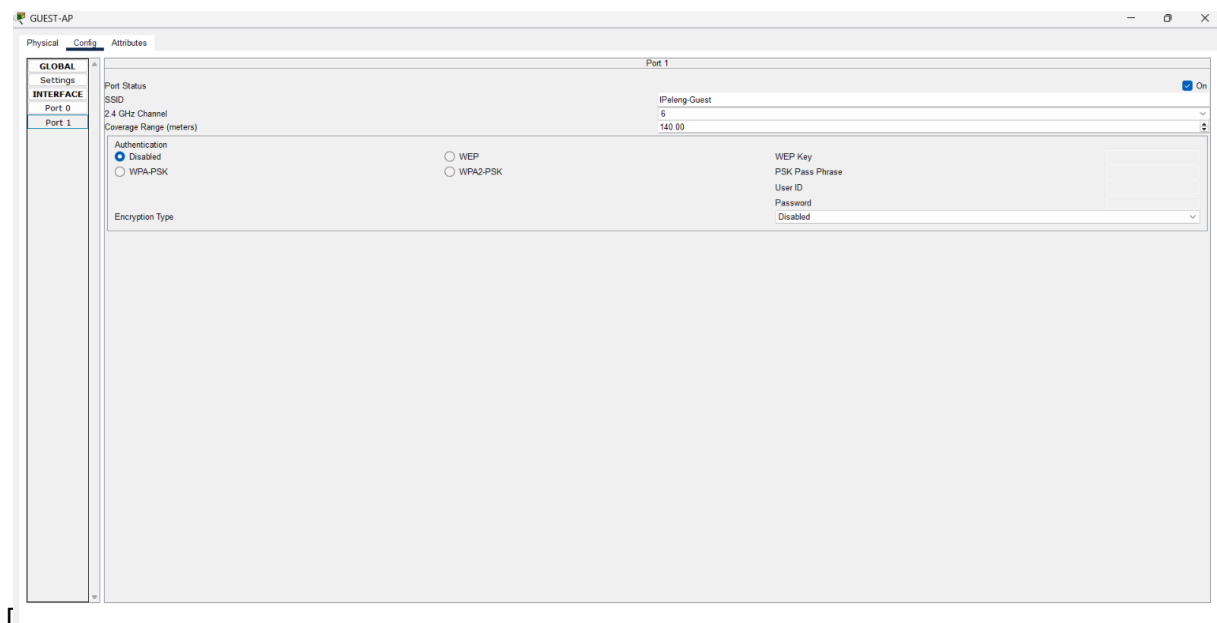


Figure 9. Guest access point and wireless client configuration.

12. Management VLAN

Device	Management IP
Core-SW	172.30.11.66
SW-1-STAFF	172.30.11.67
SW-2-STUDENTS	172.30.11.68
SW-3-GUESTS	172.30.11.69
Management Gateway	172.30.11.65

VLAN 99 provides a dedicated logical network for management interfaces. This separates management addressing from normal staff, student and guest traffic.

13. Testing and Verification

The end devices were inspected to ensure correct DHCP assignments.

- The router subinterfaces were inspected using show ip interface brief.

VLAN memberships were confirmed using show vlan brief.

Trunk operations were verified using show interfaces trunk.

Ping was used to perform default-gateway and connectivity tests.

Guest wireless connectivity was verified after configuring the access point and wireless client.

The screenshots obtained during the practical implementation serve as the evidence. Only the results that appear in the submitted screenshots should be considered as final test evidence.

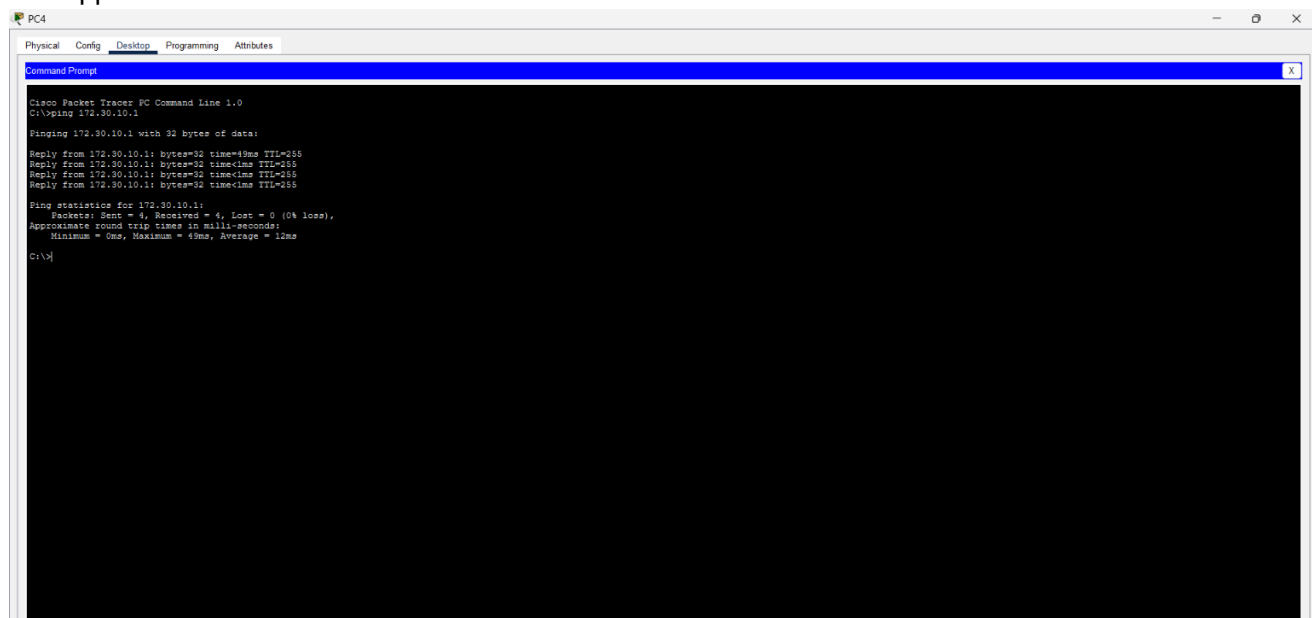


Figure 10. Connectivity/ping test evidence.

VLAN verification evidence

```
SW-1-STAFF2960 switch
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
Switch(config-if)#ip address 172.30.11.67 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#ip default-gateway 172.30.11.65
Switch(config)#

Switch con0 is now available.

Press RETURN to get started.

Switch#enable
Switch#show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7,
                                Fa0/8, Fa0/9, Fa0/10, Fa0/11,
                                Fa0/12, Fa0/13, Fa0/14, Fa0/15,
                                Fa0/16, Fa0/17, Fa0/18, Fa0/19,
                                Fa0/20, Fa0/21, Fa0/22, Fa0/23,
                                Fa0/24, Gig0/2
10   STAFF                   active    Fa0/2, Fa0/2, Fa0/3
99   VLAN099                 active
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Trunk verification evidence.

```
SW-3-GUESTS2960 switch
Physical Config CLI Attributes
IOS Command Line Interface

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
Switch(config)#interface gigabitethernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#ip default-gateway 172.30.11.65
Switch(config)#
%LINEPROTO-5-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config)#enable
% Incomplete command.
Switch(config)#show interfaces trunk
% Invalid input detected at '' marker.
Switch(config)#enable
% Incomplete command.
Switch(config)#show interfaces trunk
% Invalid input detected at '' marker.
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#enable
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
----      -
Gig0/1    on        802.1q          trunking      1
Gig0/1    Vlans allowed on trunk
Gig0/1    1-1005
Gig0/1    Vlans allowed and active in management domain
Gig0/1    1,30,99
Gig0/1    Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,30,99
Switch#
```

14. Assigned Feature: VLAN Segmentation

The main networking aspect that has been accomplished with the help of this milestone was segmentation through VLANs. VLAN 10 is set up so that it hosts staff, VLAN 20 is for students, VLAN 30 is for guests, and VLAN 99 is for management. This way, everyone's traffic is segregated on a logical level, while the organisation can still make use of the same switching infrastructure.

15. Security and Network Segmentation

The VLAN design reduces unnecessary Layer 2 traffic among the user groups and provides logical isolation between the staff, students, the guests, and the management. The VLAN level ensures that the guest network is isolated from the staff and student's access network. Other security measures, such as ACLs, firewall policies, and advanced wireless authentication features, can be considered during the next enhancement of the infrastructure. The current design also isolates the management network and ensures the logical separation that can help the management team to have greater control over the network resources.

16. Scalability

The addressing plan was developed to cater the organizations expected 25% growth rate. The current /25 and /26 subnets have spare addresses in each user segment that reduce the need for renumbering with organizational growth. The star topology also allows additional access devices to be added through the core infrastructure.

17. Problems Encountered and Solutions

Problem	Solution
Router CLI was initially in User EXEC mode.	Entered enable followed by configure terminal.
Guest device initially selected did not provide the required wireless configuration.	Replaced it with Access Point-PT.
Guest laptop initially lacked a compatible wireless interface.	Added the appropriate wireless interface/module before configuring the wireless client.
Trunk configuration required correction on the guest switch.	Re-applied switchport mode trunk on the relevant uplink.

18. Configuration Backup

The router configuration was saved using copy running-config startup-config. The switch configurations were also saved after implementation so that the completed configuration would persist after a restart.

19. Evidence Submitted

Completed Packet Tracer topology.

VLAN configuration screenshots.

Trunk configuration screenshots.

Router interface verification.

DHCP configuration and client addressing.

Printer static addressing.

Guest access point/wireless configuration.

Connectivity and ping tests.

Management VLAN addressing.

20. Conclusion

Milestone 2 implemented the network design for Ipeleng Study & Tutoring Hub in Cisco Packet Tracer. The implementation provides separate staff, student, guest and management VLANs, router-on-a-stick

inter-VLAN routing, DHCP services, static management addressing and a guest wireless network. The physical and logical design follows the requirements established in Milestone 1 and provides room for future growth without requiring immediate renumbering.

The completed configuration and collected screenshots provide practical evidence of the implementation and verification of the proposed network.

21. References

LAMOLA MC. Milestone 1: Ipeleng Study & Tutoring Hub Network Design. CMPG 325 – Computer Networks, 2026.

Cisco Packet Tracer. Network simulation and configuration environment used for the implementation.